

# SACHIN KUMAR PAL

SQL | Python | Power BI | PostgreSQL | FastAPI

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## SUMMARY

MCA student and Smart India Hackathon 2025 winner with hands-on experience in SQL, Python, PostgreSQL, and Power BI. Skilled at building end-to-end data workflows — from data generation and ETL pipelines to SQL analytics and interactive dashboards — with a strong grounding in joins, subqueries, window functions, CTEs, and relational database design. Comfortable with database schema design, query optimization, data validation, and turning raw data into business insights through dashboards and reports; also builds backend APIs using FastAPI.

## TECHNICAL SKILLS

**Data & BI:** Power BI (dashboard design, KPI/DAX basics), Pandas, ETL pipeline development, data validation & cleaning, data-driven reporting

**SQL:** Joins, Subqueries, CTEs, Window Functions, Aggregations, Stored Procedures, Query Optimization, Views

**Databases:** PostgreSQL, SQL Server, Supabase, Relational Database Design & Normalization

**Programming:** Python, SQL, Java (OOP, Core Concepts)

**Backend:** FastAPI, REST APIs

**Core Concepts:** OOP, Data Structures & Algorithms, Exception Handling

**Tools:** Git, GitHub, SQL Server Management Studio (SSMS), Linux, VS Code

## PROJECTS

**FinPulse — BNPL (Buy Now Pay Later) Data Analytics Platform** | *Python, PostgreSQL/Supabase, ETL, SQL, Power BI*

- Built an end-to-end data pipeline simulating a BNPL lending platform, generating and loading roughly 510,500 records across 6 relational tables (customers, merchants, loans, repayments, credit scores, transactions).
- Designed and implemented a Python ETL pipeline with field-level validation, duplicate prevention, foreign-key resolution, and batch inserts (500 rows/batch) into a Supabase PostgreSQL database.
- Wrote 38 SQL analysis queries and 21 database views, including an executive KPI view, to answer business questions on loan defaults, outstanding portfolio value, and repayment trends.
- Designed a 5-page Power BI dashboard (Executive Overview, Loan Analysis, Customer Analysis, Repayment & Risk, Merchant & Transactions) surfacing a ₹591 crore loan portfolio, 10.15% default rate, and ₹118 crore total repayments.
- Added automated data-quality and incremental-loading tests to verify validation rules and prevent duplicate inserts across repeated pipeline runs.

GitHub: [github.com/sachinkumar2304/FinePulse---BNPL](https://github.com/sachinkumar2304/FinePulse---BNPL)

**VaaniPath — AI-Powered Multilingual Education Platform** | *FastAPI, React, PostgreSQL/Supabase, AI Translation APIs*

- Designed the PostgreSQL/Supabase database schema and API-database integration for a platform supporting 22 Indian languages (document translation, audio translation, video dubbing).
- Contributed a fallback mechanism for the translation engine to improve reliability when the primary service fails.
- Collaborated in a team environment, focusing on database and data-flow design.

**Snackify — Full-Stack Cafe Ordering Platform** | *Next.js, React, PostgreSQL/Supabase, Vercel*

- Designed the PostgreSQL database schema for a cafe ordering system covering customers, orders, cart, and admin management.
- Built and tested core order-processing queries and admin workflows (menu management, order-status updates), using Supabase authentication and access control to protect data.
- Deployed the application on Vercel as a live, working project.

Live: [snackify-five.vercel.app](https://snackify-five.vercel.app) | GitHub: [github.com/sachinkumar2304/Cafe\\_project](https://github.com/sachinkumar2304/Cafe_project)

## ACHIEVEMENT

- Winner — Smart India Hackathon (SIH) 2025: Handled the backend and database components of the winning team project.

## EDUCATION

**Master of Computer Applications (MCA)** | *Bharti Vidyapeeth Institute of Management and Information Technology, Navi Mumbai* | 2025–2027

Semester I: 7.11 CGPA | Semester II: 7.50 CGPA

**B.Sc. Computer Science** | *CHM College, Ulhasnagar*

CGPA: 8.24